Weak Laws of Large Numbers

Stats 220A

Independence vs. uncorrelatedness

Definition

A family $\{X_i : i \in I\}$ with $\mathbb{E}X_i^2 < \infty$ is **uncorrelated** if $\mathbb{E}[X_i X_j] = \mathbb{E}[X_i]\,\mathbb{E}[X_j]$ whenever $i \neq j$.

The hierarchy.

$$\text{independent} \;\Rightarrow\; \text{uncorrelated} \;\Rightarrow\; \text{pairwise uncorrelated.}$$

Neither implication reverses in general.

Why care? The WLLN for uncorrelated variables is almost as easy as for independent ones — you only need variance additivity, and uncorrelatedness is exactly what makes that hold.

Concrete example: uncorrelated but dependent. Let $X \sim \mathrm{Uniform}(-1, 1)$ and $Y = X^2$. Then $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$, so X and Y are uncorrelated. But Y is a deterministic function of X — they are maximally dependent.

This example shows the two concepts are genuinely distinct, not just a matter of definition.

Variance additivity for uncorrelated variables

Theorem

Let $X_1, \ldots, X_n$ have $\mathbb{E}(X_i^2) < \infty$ and be uncorrelated. Then $\mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)$.

Proof.

Let $\mu_i = \mathbb{E}X_i$ and expand:

$$\mathrm{Var}(S_n) = \mathbb{E}\left(\sum_{i=1}^{n}(X_i - \mu_i)\right)^2 = \sum_{i=1}^{n}\mathbb{E}(X_i - \mu_i)^2 + 2\sum_{1\le j<i\le n}\mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]\,.$$

The first sum is $\sum_i \mathrm{Var}(X_i)$. For each cross term: $\mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \mathbb{E}[X_iX_j] - \mu_i\mu_j = 0$ by uncorrelatedness. All cross terms vanish. $\square$

Remark

For *correlated* variables, the cross term is $\mathrm{Cov}(X_i, X_j)$ and may be nonzero. The WLLN can still hold if covariances decay fast enough, but the proof requires more care (Exercise 2.2.2 in Durrett).

L^2 Weak Law of Large Numbers

Theorem (L^2 WLLN)

Let $X_1, X_2, \ldots$ be uncorrelated with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) \leq C < \infty$. Let $S_n = X_1 + \cdots + X_n$. Then as $n \to \infty$:

$$\frac{S_n}{n} \xrightarrow{L^2} \mu \qquad \text{and} \qquad \frac{S_n}{n} \xrightarrow{\mathbb{P}} \mu.$$

Proof.

Since $\mathbb{E}(S_n/n) = \mu$, the L^2 error is:

$$\mathbb{E}\left[\left(\frac{S_n}{n} - \mu\right)^2\right] = \mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{\mathrm{Var}(S_n)}{n^2} = \frac{\sum_{i=1}^n \mathrm{Var}(X_i)}{n^2} \leq \frac{Cn}{n^2} = \frac{C}{n} \to 0.$$

This gives L^2 convergence. Since $L^2 \Rightarrow$ in probability (Lecture 1), we also get $S_n/n \xrightarrow{\mathbb{P}} \mu$. $\square$

Unpacking the proof

The proof reduces to one chain of inequalities:

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \varepsilon\right) \leq \frac{\mathrm{Var}(S_n/n)}{\varepsilon^2} \leq \frac{C}{n\varepsilon^2} \to 0.$$

Three ingredients, each doing exactly one job:

1. **Uncorrelatedness** $\Rightarrow \mathrm{Var}(S_n) = \sum_i \mathrm{Var}(X_i)$ (no cross terms).
2. **Uniform bound** $\mathrm{Var}(X_i) \leq C \Rightarrow \mathrm{Var}(S_n) \leq Cn$.
3. **Chebyshev** $\Rightarrow \mathrm{Var}(S_n/n) = C/n$ translates into a probability bound.

Rate. The bound $C/(n\varepsilon^2)$ is the right order: for i.i.d. variables, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$ exactly.

What we do not need. We never assumed finite third or fourth moment. Finite variance (uniformly bounded) is the sharp condition for this approach. We can also relax $\mathrm{Var}(X_i) \leq C$ to $\mathrm{Var}(X_i)/i \to 0$; see Exercise 2.2.1 in Durrett.

Application: Bernstein polynomial approximation

Theorem (Weierstrass, proved probabilistically)

Let f be continuous on $[0,1]$ and let $B_n(f;\,x) = \sum_{m=0}^{n} \binom{n}{m} x^m (1-x)^{n-m} f(m/n)$. Then $\sup_{x\in[0,1]} |B_n(f;\,x) - f(x)| \to 0$ as $n \to \infty$.

Probabilistic interpretation. If $S_n \sim \mathrm{Binomial}(n, x)$ then $B_n(f;\,x) = \mathbb{E}[f(S_n/n)]$. The WLLN says $S_n/n \xrightarrow{\mathbb{P}} x$; since f is continuous we expect $f(S_n/n) \xrightarrow{\mathbb{P}} f(x)$.

Proof sketch.

Fix $\varepsilon > 0$; by uniform continuity on $[0,1]$ pick δ so $|u - v| < \delta \Rightarrow |f(u) - f(v)| < \varepsilon/2$. Let $M = \sup |f|$. Then:

$$
\begin{aligned}
|B_n(f;\,x) - f(x)| &\le \mathbb{E}|f(S_n/n) - f(x)| \\
&\le \tfrac{\varepsilon}{2} + 2M\,\mathbb{P}(|S_n/n - x| \ge \delta) \le \tfrac{\varepsilon}{2} + 2M \cdot \frac{x(1-x)}{n\delta^2} \le \tfrac{\varepsilon}{2} + \frac{M}{2n\delta^2}.
\end{aligned}
$$

Since $x(1-x) \le 1/4$ the bound is *uniform in x*. For $n \ge M/(\varepsilon\delta^2)$ the right side is $< \varepsilon$. $\square$

Why this is remarkable

What we just proved: every continuous function on $[0, 1]$ is a *uniform* limit of polynomials. This is the Weierstrass approximation theorem — and we proved it using probability.

Two things to notice:

1. The probabilistic rate of convergence becomes the analytic rate of approximation. The Chebyshev bound $x(1-x)/n \leq 1/(4n)$ controls both the WLLN and the approximation error uniformly. The algebraic and probabilistic calculations are the same calculation.

2. We used the proof of the WLLN, not just its conclusion. The result $S_n/n \xrightarrow{\mathbb{P}} x$ would only give pointwise convergence $f(S_n/n) \xrightarrow{\mathbb{P}} f(x)$. Uniformity in x requires the explicit rate $\mathbb{P}(|S_n/n - x| > \delta) \leq x(1-x)/(n\delta^2) \leq 1/(4n\delta^2)$, which is uniform in x precisely because $\mathrm{Var}(\mathrm{Bernoulli}(x)) = x(1-x) \leq 1/4$.

Lesson. The WLLN is not just an asymptotic result about averages. Quantitative versions give explicit function approximation bounds.

Triangular arrays: setup and motivation

In many limit theorems the random variables *depend on* n — the distribution of each term changes as the number of terms grows.

Definition

A **triangular array** is a collection $\{X_{n,k} : 1 \le k \le n\}$ where for each n, the row $X_{n,1}, \ldots, X_{n,n}$ consists of independent random variables. The **row sum** is $S_n = \sum_{k=1}^{n} X_{n,k}$.

Examples.

- *Coupon collector:* W_k = draws to get the k-th new coupon; success probability $(n-k+1)/n$ depends on k and n.
- *Cycle indicators:* $X_{n,k} = \mathbf{1}(\text{position } k \text{ closes a cycle})$ in a random permutation of $\{1, \ldots, n\}$.
- *Binomial*$(n, \lambda/n)$*:* each term has success probability $\lambda/n \to 0$.

A single i.i.d. sequence $X_1, X_2, \ldots$ is the degenerate case $X_{n,k} = X_k$.

General triangular array WLLN

Theorem

Let $\mu_n = \mathbb{E}S_n$ *and* $\sigma_n^2 = \mathrm{Var}(S_n)$. *If* $\sigma_n^2/b_n^2 \to 0$ *for some* $b_n \to \infty$, *then*

$$\frac{S_n - \mu_n}{b_n} \xrightarrow{\mathbb{P}} 0.$$

Proof.

Chebyshev directly: $\mathbb{P}(|S_n - \mu_n|/b_n > \varepsilon) \leq \mathrm{Var}(S_n)/(b_n^2\varepsilon^2) = \sigma_n^2/(b_n^2\varepsilon^2) \to 0.$ □

Template for applications. Every application follows the same three steps:

1. Compute $\mu_n = \mathbb{E}S_n$ and bound $\sigma_n^2 = \mathrm{Var}(S_n)$.
2. Choose b_n so that $\sigma_n^2/b_n^2 \to 0$ and $b_n \to \infty$.
3. Conclude $S_n \approx \mu_n$ in probability at scale b_n.

The theorem is trivial. The substance is in the calculations for each example.

A useful estimate: partial harmonic sums

In both applications, the key sums are harmonic. Recall:

$$\log n \;\le\; \sum_{m=1}^{n} \frac{1}{m} \;\le\; 1 + \log n, \qquad \sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{6} < \infty.$$

The first follows from comparing $\sum 1/m$ with $\int dx/x$:

$$\sum_{m=1}^{n} \frac{1}{m} \ge \int_1^{n+1} \frac{dx}{x} = \log(n+1) \ge \log n, \qquad \sum_{m=1}^{n} \frac{1}{m} \le 1 + \int_1^{n} \frac{dx}{x} = 1 + \log n.$$

Why this matters. Both the coupon collector expected time and the expected number of cycles are exactly $\sum_{m=1}^{n} 1/m \sim \log n$. The same harmonic sum appears because both problems are waiting for rare events at geometrically spaced probabilities.

Application: coupon collector's problem

Setup. Draw i.i.d. Uniform$\{1, \ldots, n\}$ samples. Let T_n = number of draws to see all n values.
Write $T_n = \sum_{k=1}^{n} W_k$ where W_k = additional draws needed once $k-1$ distinct values are collected. Then $W_k \sim \mathrm{Geometric}\left(\frac{n-k+1}{n}\right)$, independent across k.

Mean.

$$\mathbb{E}[T_n] = \sum_{k=1}^{n} \frac{n}{n-k+1} = n \sum_{m=1}^{n} \frac{1}{m} \sim n \log n.$$

Variance. Since $\mathrm{Var}(\mathrm{Geom}(p)) = (1-p)/p^2 \leq 1/p^2$:

$$\mathrm{Var}(T_n) \leq \sum_{k=1}^{n} \left(\frac{n}{n-k+1}\right)^2 = n^2 \sum_{m=1}^{n} \frac{1}{m^2} \leq \frac{\pi^2 n^2}{6}.$$

Conclusion. Take $b_n = n \log n$. Then
$\mathrm{Var}(T_n)/b_n^2 = O(n^2/(n \log n)^2) = O(1/(\log n)^2) \to 0$, so by the theorem:

$$\frac{T_n}{n \log n} \xrightarrow{\mathbb{P}} 1.$$

Coupon collector: what the rate means

The answer. To collect all n coupons, you need about $n \log n$ draws.

Concrete example. For $n = 365$ (birthdays): $365 \cdot \log(365) \approx 2153$ people needed to see every birthday.

Why $n \log n$ and not just n?
The first coupon is free ($W_1 = 1$ always). Each subsequent coupon takes slightly longer. The k-th new coupon requires $\mathbb{E}W_k = n/(n-k+1)$ draws. When most coupons are collected (k close to n), finding the last few takes $O(n)$ draws each. Summing the waiting times:

$$n \cdot \frac{1}{n} + n \cdot \frac{1}{n-1} + \cdots + n \cdot \frac{1}{1} = n\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right) \approx n \log n.$$

The last coupon dominates. The final missing item takes $\mathbb{E}W_n = n$ draws alone — contributing a full factor of n to the total. The harmonic sum $\log n$ is exactly the overhead of collecting the "tail" of rare coupons.

Application: cycle structure of random permutations

Setup. Let π be Uniform over all $n!$ permutations of $\{1, \ldots, n\}$. Every permutation decomposes uniquely into cycles: e.g., $(1\,3\,6)(2\,9\,7\,5)(4\,8)$ has 3 cycles. Let S_n = number of cycles of π.

Lemma

Define $X_{n,k} = 1$ if position k closes a cycle in the canonical cycle decomposition, 0 otherwise. Then $X_{n,1}, \ldots, X_{n,n}$ are **independent** *with $\mathbb{P}(X_{n,k} = 1) = 1/(n-k+1)$.*

Intuition. When assigning $\pi(i_k)$ at step k, there are $n-k+1$ remaining values; exactly one closes the current cycle.

Mean and variance.

$$\mathbb{E}S_n = \sum_{k=1}^{n} \frac{1}{n-k+1} = \sum_{m=1}^{n} \frac{1}{m} \sim \log n.$$

Since $X_{n,k}^2 = X_{n,k}$: $\mathrm{Var}(X_{n,k}) \le \mathbb{E}X_{n,k} = 1/(n-k+1)$. So $\mathrm{Var}(S_n) = \sum_k \mathrm{Var}(X_{n,k}) \le \mathbb{E}S_n \sim \log n$. $\square$

Taking $b_n = (\log n)^{0.5+\varepsilon}$ gives $\mathrm{Var}(S_n)/b_n^2 \le (\log n)/(\log n)^{1+2\varepsilon} \to 0$. Therefore $S_n/\log n \xrightarrow{\mathbb{P}} 1$.

Random permutations: what it means

Result. A random permutation of n elements has approximately $\log n$ cycles.

Examples.

- $n = 100$: roughly $\log 100 \approx 4.6$ cycles.
- $n = 10^6$: roughly $\log(10^6) \approx 13.8$ cycles.
- $n = 10^{100}$: roughly $\log(10^{100}) \approx 230$ cycles.

Cycles grow *logarithmically* — extremely slowly.

The same harmonic sum appears everywhere.

$$\mathbb{E}T_n(\text{coupon}) = n\sum_{m=1}^{n}\frac{1}{m}, \qquad \mathbb{E}S_n(\text{cycles}) = \sum_{m=1}^{n}\frac{1}{m}, \qquad \mathbb{E}R_n(\text{records}) = \sum_{m=1}^{n}\frac{1}{m}.$$

These three problems — coupon collector, cycle counts, record values (Lecture 1) — all produce $\sum 1/m \sim \log n$ because each is counting events that occur with probability $1/m$ at step m, independently.

Note. The a.s. version $S_n/\log n \xrightarrow{\text{a.s.}} 1$ is also true (proved in Lecture 1 via BC2). The WLLN is weaker but requires much less work.

Why we need truncation

The L^2 WLLN requires $\mathrm{Var}(X_i) \le C < \infty$. This is far stronger than necessary.

Can we prove $S_n/n \xrightarrow{\mathbb{P}} \mu$ under just $\mathbb{E}|X_i| < \infty$?
The Chebyshev approach breaks down when $\mathbb{E}X^2 = \infty$: we cannot bound $\mathrm{Var}(S_n)$.

The truncation idea. Replace each X_i with a *trimmed* version:

$$\bar{X}_i^{(M)} = X_i \cdot \mathbf{1}_{\{|X_i| \le M\}}.$$

The trimmed variable satisfies $|\bar{X}_i^{(M)}| \le M$, so it has finite variance $\le M^2$. We then show three things:

1. *Few terms are trimmed:* $\mathbb{P}(X_i \ne \bar{X}_i^{(M)}) = \mathbb{P}(|X_i| > M)$ is small if M is large.
2. *Trimmed sum concentrates:* L^2 WLLN applies to $\bar{X}_i^{(M)}$ for each fixed M.
3. *Trimmed mean approaches μ:* $\mathbb{E}[\bar{X}^{(M)}] \to \mu$ as $M \to \infty$.

This is the central technical idea for handling infinite-variance distributions.

The layer-cake formula

Lemma

For $Y \geq 0$ and $p > 0$: $\mathbb{E}[Y^p] = \int_0^\infty p\, y^{p-1}\, \mathbb{P}(Y > y)\, dy$.

Proof.

Fubini (all terms ≥ 0):

$\int_0^\infty p\, y^{p-1} \mathbb{P}(Y > y)\, dy = \int_\Omega \int_0^{Y(\omega)} p\, y^{p-1}\, dy\, d\mathbb{P}(\omega) = \int_\Omega Y(\omega)^p\, d\mathbb{P}(\omega) = \mathbb{E}[Y^p]$. □

Two key uses.

$p = 1$: $\mathbb{E}|X| = \int_0^\infty \mathbb{P}(|X| > y)\, dy$. Mean is finite $\Leftrightarrow$ tail CDF is integrable.

$p = 2$, *truncated:* For $\bar{X} = X\mathbf{1}_{|X| \leq n}$: $\mathbb{E}[\bar{X}^2] \leq \int_0^n 2y\, \mathbb{P}(|X| > y)\, dy$. This bounds the variance of the trimmed variable using only the tail of $|X|$ — no second moment required.

Why the truncated bound works. $\mathbb{P}(|\bar{X}| > y) = \mathbb{P}(|X| > y) - \mathbb{P}(|X| > n) \leq \mathbb{P}(|X| > y)$ for $y \leq n$, and $= 0$ for $y > n$. Integrating by layers gives the bound.

Triangular array WLLN with truncation

Theorem

For each n, let $X_{n,1}, \ldots, X_{n,n}$ be independent. Let $b_n \to \infty$ and $\bar{X}_{n,k} = X_{n,k}\mathbf{1}_{\{|X_{n,k}| \le b_n\}}$. If:

(i) $\displaystyle\sum_{k=1}^{n} \mathbb{P}(|X_{n,k}| > b_n) \to 0$, *and*

(ii) $\displaystyle b_n^{-2} \sum_{k=1}^{n} \mathbb{E}[\bar{X}_{n,k}^2] \to 0$,

then with $a_n = \sum_{k=1}^{n} \mathbb{E}[\bar{X}_{n,k}]$ and $S_n = \sum_k X_{n,k}$:

$$\frac{S_n - a_n}{b_n} \xrightarrow{\mathbb{P}} 0.$$

Condition (i) ensures the original and truncated sums agree with high probability.
Condition (ii) ensures the truncated sum concentrates around its mean.

Proof of the truncation theorem

Let $\bar{S}_n = \sum_k \bar{X}_{n,k}$. For any $\varepsilon > 0$, split:

$$\mathbb{P}\left(\left|\frac{S_n - a_n}{b_n}\right| > \varepsilon\right) \leq \underbrace{\mathbb{P}(S_n \neq \bar{S}_n)}_{\text{truncation error}} + \underbrace{\mathbb{P}\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right)}_{\text{trimmed sum}}.$$

Truncation error. $\{S_n \neq \bar{S}_n\} \subseteq \bigcup_k \{|X_{n,k}| > b_n\}$, so by a union bound:

$$\mathbb{P}(S_n \neq \bar{S}_n) \leq \sum_{k=1}^{n} \mathbb{P}(|X_{n,k}| > b_n) \to 0 \quad \text{by (i).}$$

Trimmed sum. Since $a_n = \mathbb{E}[\bar{S}_n]$, Chebyshev and variance additivity give:

$$\mathbb{P}\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right) \leq \frac{\mathrm{Var}(\bar{S}_n)}{b_n^2 \varepsilon^2} = \frac{\sum_k \mathrm{Var}(\bar{X}_{n,k})}{b_n^2 \varepsilon^2} \leq \frac{\sum_k \mathbb{E}[\bar{X}_{n,k}^2]}{b_n^2 \varepsilon^2} \to 0 \quad \text{by (ii).}$$

The minimal WLLN: necessary and sufficient condition

Theorem

Let $X_1, X_2, \ldots$ be i.i.d. with $x\,\mathbb{P}(|X_1| > x) \to 0$ as $x \to \infty$, and $\mu_n = \mathbb{E}[X_1 \mathbf{1}_{\{|X_1| \le n\}}]$. Then $S_n/n - \mu_n \xrightarrow{\mathbb{P}} 0$.

Proof.

Apply the truncation theorem with $X_{n,k} = X_k$ and $b_n = n$.
Condition (i): $\sum_k \mathbb{P}(|X_k| > n) = n\mathbb{P}(|X_1| > n) \to 0$ by hypothesis.
Condition (ii): By the layer-cake bound $\mathbb{E}[\bar{X}_{n,1}^2] \le \int_0^n 2y\mathbb{P}(|X_1| > y)\,dy$:

$$\frac{1}{n}\mathbb{E}[\bar{X}_{n,1}^2] \le \frac{1}{n}\int_0^n 2y\mathbb{P}(|X_1| > y)\,dy.$$

The integrand $g(y) = 2y\mathbb{P}(|X_1| > y) \to 0$ (hypothesis), and g is bounded, so its average over $[0, n]$ goes to 0. Thus $b_n^{-2}\sum_k \mathbb{E}[\bar{X}_{n,k}^2] = n^{-1}\mathbb{E}[\bar{X}_{n,1}^2] \to 0$. $\square$

Remark

The condition $x\mathbb{P}(|X| > x) \to 0$ is *necessary* for constants a_n to exist with $S_n/n - a_n \xrightarrow{\mathbb{P}} 0$.

WLLN under $\mathbb{E}|X| < \infty$

Theorem (Weak Law of Large Numbers)

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}|X_i| < \infty$ and $\mathbb{E}X_i = \mu$. Then $S_n/n \xrightarrow{\mathbb{P}} \mu$.

Proof.

Tail condition. By dominated convergence, since $|X_1| \in L^1$ and $\mathbf{1}_{\{|X_1|>x\}} \to 0$ a.s.: $x\,\mathbb{P}(|X_1| > x) \le \mathbb{E}\big[|X_1|\mathbf{1}_{\{|X_1|>x\}}\big] \to 0$ as $x \to \infty$.
Centering. $\mu_n = \mathbb{E}[X_1\mathbf{1}_{\{|X_1|\le n\}}] \to \mathbb{E}[X_1] = \mu$ by dominated convergence, since $X_1\mathbf{1}_{\{|X_1|\le n\}} \to X_1$ a.s. and $|X_1\mathbf{1}_{\{|X_1|\le n\}}| \le |X_1| \in L^1$.
Combine. The previous theorem gives $S_n/n - \mu_n \xrightarrow{\mathbb{P}} 0$. Since $\mu_n \to \mu$:

$$\frac{S_n}{n} = \left(\frac{S_n}{n} - \mu_n\right) + \mu_n \xrightarrow{\mathbb{P}} 0 + \mu = \mu. \qquad \square$$

Remark. The proof has two clean parts: (1) verify $\mathbb{E}|X| < \infty$ implies the tail condition; (2) dominated convergence handles the centering. The heavy lifting is all in the truncation theorem.

The sharp condition: necessity of $x\mathbb{P}(|X| > x) \to 0$

Remark

The condition $x\mathbb{P}(|X| > x) \to 0$ is *necessary* for the existence of *any* constants a_n with $S_n/n - a_n \xrightarrow{\mathbb{P}} 0$.

Implication for the hierarchy.

Condition	Conclusion
$\mathrm{Var}(X_i) \leq C$ + uncorrelated	$S_n/n \xrightarrow{L^2} \mu$ at rate C/n
$\mathbb{E}\lvert X_i\rvert < \infty$, i.i.d.	$S_n/n \xrightarrow{\mathbb{P}} \mu$
$x\mathbb{P}(\lvert X\rvert > x) \to 0$, i.i.d.	$S_n/n - \mu_n \xrightarrow{\mathbb{P}} 0$ (centering grows)
Neither condition	No weak law

When the centering grows. If $\mathbb{P}(|X| > x) \sim c/x^{1+\alpha}$ with $0 < \alpha < 1$: $\mathbb{E}|X| = \infty$ but $x\mathbb{P}(|X| > x) \to 0$, and $\mu_n \sim cn^{1-\alpha}/(1-\alpha) \to \infty$. A WLLN holds but with growing recentering.

If $\mathbb{P}(|X| > x) \sim c/x$ (Cauchy): $x\mathbb{P}(|X| > x) \to c \neq 0$. The tail condition fails. No WLLN at any scale.

The St. Petersburg game

Rules. Flip a fair coin until the first head. If the first head appears on toss j, you win 2^j dollars. The distribution: $\mathbb{P}(X = 2^j) = 2^{-j}$ for $j = 1, 2, 3, \ldots$
The paradox. $\mathbb{E}X = \sum_{j=1}^{\infty} 2^j \cdot 2^{-j} = \infty$. A risk-neutral player should pay *any finite amount* to play. Yet no one would pay \$100 per game.
The mathematical question. If $S_n = X_1 + \cdots + X_n$ (n i.i.d. games), what is the right normalization? There is no finite μ to converge to. Does a law of large numbers hold at all?
Historical context. Posed by Nicolas Bernoulli (1713). It motivated utility theory (Daniel Bernoulli, 1738) and occupied mathematicians for two centuries. The correct probabilistic resolution — a WLLN with growing centering — was given by Feller (1945).

Spoiler. The answer is $S_n/(n \log_2 n) \xrightarrow{\mathbb{P}} 1$: the fair price per game is $\log_2 n$ dollars, growing slowly with the number of games played.

Setting up the truncation for St. Petersburg

Apply the triangular array theorem with $X_{n,k} = X_k$ i.i.d. St. Petersburg, and choose b_n carefully.
Key tail calculation. For integer m:

$$\mathbb{P}(X \geq 2^m) = \sum_{j=m}^{\infty} 2^{-j} = 2^{-m+1}.$$

Choice of b_n. Condition (i) requires $n\mathbb{P}(X > b_n) \to 0$. Set $m(n) = \lfloor \log_2 n \rfloor + K(n)$ where $K(n) \to \infty$ (e.g. $K(n) = \lfloor \log_2 \log_2 n \rfloor$), and $b_n = 2^{m(n)}$. Then:

$$n\mathbb{P}(X \geq b_n) = n \cdot 2^{-m(n)+1} = 2n/b_n = 2 \cdot 2^{-K(n)} \to 0. \checkmark$$

Checking condition (ii).

$$\mathbb{E}[\bar{X}_{n,1}^2] = \sum_{j=1}^{m(n)} 2^{2j} \cdot 2^{-j} = \sum_{j=1}^{m(n)} 2^j \leq 2^{m(n)+1} = 2b_n.$$

So $b_n^{-2} \cdot n \cdot \mathbb{E}[\bar{X}_{n,1}^2] \leq 2n/b_n = 2 \cdot 2^{-K(n)} \to 0$. $\checkmark$

St. Petersburg: the WLLN and its interpretation

Computing the centering.

$$a_n = n\,\mathbb{E}[\bar{X}_{n,1}] = n \sum_{j=1}^{m(n)} 2^j \cdot 2^{-j} = n \cdot m(n).$$

Since $m(n) = \log_2 n + K(n)$ and $K(n)/\log_2 n \to 0$: $a_n/(n \log_2 n) = m(n)/\log_2 n \to 1$.

Applying the theorem. $(S_n - a_n)/b_n \xrightarrow{\mathbb{P}} 0$ and $b_n = n \cdot 2^{K(n)} \ll n \log_2 n$, giving:

$$\boxed{\frac{S_n}{n \log_2 n} \xrightarrow{\mathbb{P}} 1.}$$

Interpretation. After n games, total winnings $\approx n \log_2 n$, so the fair price per game is $\log_2 n$ dollars.

n	$\log_2 n$	Fair price per play
1024	10	\$10
10^6	20	\$20
10^{12}	40	\$40

Nicolas Bernoulli's suggestion of 40 ducats corresponds to $\approx 10^{12}$ games.

Summary

The WLLN landscape:

Condition	**Result**
$\mathrm{Var}(X_i) \leq C$, uncorrelated	$S_n/n \xrightarrow{L^2} \mu$ at rate C/n
$\mathbb{E}\lvert X_i\rvert < \infty$, i.i.d.	$S_n/n \xrightarrow{\mathbb{P}} \mu$
$x\mathbb{P}(\lvert X\rvert > x) \to 0$, i.i.d.	$S_n/n - \mu_n \xrightarrow{\mathbb{P}} 0$, centering grows
St. Petersburg ($\mathbb{E}X = \infty$)	$S_n/(n \log_2 n) \xrightarrow{\mathbb{P}} 1$
Cauchy	No WLLN at any scale